Subject: Bt605 (past papers)

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1) **Define Genetic modification?**

   Genetic modification: any alteration of genetic material, capable of producing new substances, performing new functions

2) **What are Design babies?**

   A designer baby is a term used by journalists to describe this frightening scenario. Designer babies are children-genetically engineered in the womb to have desired

qualities. They are produced through in vitro fertilization. First embryo is removed manipulated for desired qualities and is then placed back into mother's womb. The major disadvantage is that it is expensive not 100% save

3) **What are injuries how it can affect a researcher dealing with animal?**
All occupational injuries, including exposures to human pathogens, should be reported to the medical support services provider Eye injuries are among the most preventable types of laboratory accidents. Glasses routinely worn for vision correction do not provide the appropriate level of protection for work with hazardous materials
4) **What are risks for researchers who work with animals 2**
human health, biodiversity, animal welfare, poor communities
5) **Define cybernetics? 2**
Cybernetics has the ability to explore regulatory system, their structures and functions. It is derived from a Greek word that means governance Cybernetics is the study of interactions between man, machine and animals. Today because of this technique we can create Superhumans. This transforms the way we practice medicine, transmit thoughts and communicate with one another. Softwares are built by cybernetics to read signals from the nervous systemto record and condition the data for retransmission
6) **What is Legal status of fetus 2**
The maternal-fetal relationship is the subject of intense public policy debates. Lisa Harris and Lynn Paltrow report on courts' opinions regarding the prosecution of pregnant women for potential fetal harm. Nathan Stormer explores how contemporary biomedical images of the fetus are changing public perception of the pregnant body and fetus. Amy Salisbury and colleagues describe how researchers are attempting to measure maternal-fetal attachment and its possible implications for fetal and maternal health during pregnancy.
fetus has the right—begin his life with sound body and mind
charges of fetal abuse
refusal of hospitalization, intrauterine transfusion or surgical delivery
7) **Describe mean attitude?**
Attitudes to science: According to Wittgenstein **"scientific terms are interpreted in social context".** This has greater contribution to the economy growth. Post modernism says that results are not experimentally built but socially constructed.Science and technology are as central as ever. Keep in mind science is not done by robots actually public define science, think about research and policies
8) **What Chemical agents used as disinfectant 2**
glutaraldehyde, formaldehyde, chlorine
9) **What is the state of 21st century 2**

In the field of human procreation, the problems raised by bioethics are multiple and complex. First, there is the request from couples and individuals with regards to the

management of procreation, not only the negative requests, be it contraception or abortion, but also the positive requests concerning medically assisted procreation. Then there is the practitioner's response to the request, particularly for assisted procreation. The practitioner is at the heart of conflicting interests between the needs or desires of the patients and the reluctance, legitimate or not, of society, with the implication of a third person, embryo, fetus or newborn. And here again we are confronted with the aims of researchers wishing to obtain information concerning the efficacy, possibilities of application and potential dangers. The legists and magistrates are concerned with the legal status of the embryo and the fetus. However, if the fact of giving a legal status can solve many legal problems, it does not solve the problem of the eventual use of an embryo, and of its interpretation by philosophers and moralists. Moreover, public opinion should not be forgotten, including the attraction of rapid anathema, without recourse. Finally, there is also the role of the legislator who, here again, is confronted with several controversial conceptions.

10)Explain two types of sterilization with example 3

Removal of microbes including bacterial spores, disinfection-killing many microbes • containment level (to BSL-4)

BSL-1 this level is suitable for work involving well-characterized agents not known to consistently cause disease in healthy adult humans, and of minimal potential hazard to laboratory personnel and the environment

BSL-2 this level is similar to Biosafety Level 1 and is suitable for work involving agents of moderate potential hazard to personnel and the environment. It includes various bacteria and viruses that cause only mild disease to humans, or are difficult to contract via aerosol in a lab setting, such as C. difficile, most Chlamydiae, hepatitis A, B, and C, human immunodeficiency virus(HIV), orthopoxviruses

11)What are rights and responsibilities of patient for refusal treatment 3

Refusal: patient has a right to decline treatment unethical to force the patient patient must understand the consequences of refusal

Or

physicians must ultimately accept the patient's decision because patients have the right to decline treatment due to their autonomy, or freedom to decide. it would be unacceptable and even inhumane if patients were physically and forcefully treated against their will. A patient who refuses treatment must first be well-informed to make the decision, possess proper decision-making capacity to understand the consequences of refusal, and make the decision without manipulation or coercion by outsiders. Patients can refuse all types, even ones that are lifesaving with minimal side-effects

12)Write Ethical concern of genetic screening 3

- religious groups
- expensive/emotional distress
- done for common diseases/part of medical record???\

- deny employment, social services and insurance benefits

13)What is Maternal fatal relationship 3

Maternal-fetal relationship: The maternal-fetal relationship is the subject of intense public policy debates. Lisa Harris and Lynn Paltrow report on courts' opinions regarding the prosecution of pregnant women for potential fetal harm. Nathan Stormer explores how contemporary biomedical images of the fetus are changing public perception of the pregnant body and fetus. Amy Salisbury and colleagues describe how researchers are attempting to measure maternal-fetal attachment and its possible implications for fetal and maternal health during pregnancy.Although fundamental and seemingly unalterable, the definition of the maternal-fetal relationship is undergoing a rapid evolution of context, in part due to advances in imaging, prenatal diagnostics, genetic screening, and fetal surgery. While the formal principles of autonomy and beneficence provide an initial point from which to analyze the sometimes conflicting needs of the pregnant woman and her fetus, the maternal-fetal relationship is becoming too complex to rely on simple algorithms. As medical technology and innovation advance, and as cultural and political dialogues continue to influence the patient-physician encounter, physicians' knowledge of and participation in ethical dialogue on the maternal-fetal relationship will continue to be an increasingly important part of obstetric care.

14)What are Ethical issue of treatment of animal in lab

animal welfare and moral community, sentiency, speciesism, religious concerns
Ethical issues, including concerns for animal welfare, can arise at all stages in the generation and life span of an individual genetically engineered animal. The following sections detail some of the issues that have arisen during the peer-driven guidelines development process and associated impact analysis consultations carried out by the CCAC. The CCAC works to an accepted ethic of animal use in science, which includes the principles of the Three Rs (Reduction of animal numbers, Refinement of practices and husbandry to minimize pain and distress, and Replacement of animals with non-animal alternatives wherever possible). Together the Three Rs aim to minimize any pain and distress experienced by the animals used, and as such, they are considered the principles of humane experimental technique. However, despite the steps taken to minimize pain and distress, there is evidence of public concerns that go beyond the Three Rs and animal welfare regarding the creation and use of genetically engineered animals

Or

The ethics of animal research come into play, where many, such as the moral philosopher Peter Singer, believe that all animals are created equal; suggesting animal testing in science should be completely eliminated. The possibilities of unforeseen health risks in cloned organisms and potential negative effects of decreased genetic variation on the human gene pool are seen as ethical causes for concern in addition to the mixed ethical and social consideration of increasing

population sizes when worldwide resource availability is a problem. The social issues of cloning tend to focus on human clones in terms of both availability of cloning technology and integration of clones into society. Reproductive cloning raises the question of cost and who should have access. However, the biggest social argument is that cloning negates a person's right to individuality and ignores the potential psychological effects of such a parentless and de-individualized identity. Legally, funding has always been a concern for cloning research. Many believe the government and taxpayer money should not support research not agreed upon by a clear majority and in this respect, the U.S. Congress has continued to prohibit use of taxpayer dollars for any research that may result in the death of human embryos. However, reproductive animal cloning continues not just in the U.S., but around the world. The biggest legal issues concerning animal clones are who should be responsible for and at what depth there should be oversight and accountability, as well as the legal right to patent live organisms. (copied from handouts, you can minimize the material according to your ease)

15)How we make decisions? Explain it.

Decisions:

• long and complex history

• ethics is about decisions and making choices

• our daily conversation has an ethical component

Virtue ethics: • What is most virtuous?

• expression of the individuals than keeping the rules • we become virtuous by practicising virtue • What is most virtuous? • expression of the individuals than keeping the rules we become virtuous by practicising virtue

16)Write down equipment name used in lab? 2

Borosilicate glassware, corks for sealing organic solvents, thermometer-stirring device, thermometer bulb-heat. Laboratory coat, shoe covers, gloves, and possible respiratory protection (you can mention others also)

17)What is zoonosis? What is Zionistic disease and write name of diseases? 4

It is a disease which can be transmitted to humans from animals. Asthma and allergie re examples of **zoonosis.**

18)Write containment zone? 3

Containment: Classification of organisms according to risk group is not meant to establish the actual handling of biological hazards in the laboratory setting. For example, the risk group system does not take into account the procedures that are to be employed during the manipulation of a particular organism. Containment levels are selected to provide the end-user with a description of the minimum containment required for handling the organism safely in a laboratory setting. the containment system includes the engineering, operational, technical and physical requirements for manipulating a particular pathogen. These containment levels are applicable to

facilities such as diagnostic, research, clinical, teaching and production facilities that are working at a laboratory scale.Four containment levels are described as follows:

Containment Level 1 (CL1)

Containment Level 2 (CL2)

Containment Level 3 (CL3)

Containment Level 4 (CL4

19)What are Element of accident report

- Accident report must be prepared-five days of accident
- What is accident?
- Person involved
- Witness to the accident

20)What are Advantages of uv in sterilization

- readily available
- no known toxic residuals
- short contact time
- equipment is easy to operate and maintain

21)Enlist four discipline which shows the relationship of human with nature

Anthropocentrism: • approach centered to human beings

• rest of the nature is being there for the good of humans

• Human greediness-ecological footprints-massive alteration in nature balance

Ecocentrism**:** Centered—ecosystem

• Includes soil, water, air, forest, mountains

• biotic components are dependent on non-living

• Human ill treatment---altering nature—threatening for humans

• humans are responsible for all biological life

• ability of thinking and perceiving world as a whole

• Rolston---ecosystem is much more than the sum of its parts

Biocentrism**:** centered—biosphere

• humans are one of many millions of species

• doesn't prevent humans--using natural resources if other living organisms doesn't matter—consequentalism

Theocentrism**:** God centered approach to the world

• Comes from the religious faith---God is a creator

• Environment—belongs to God

22)Write constituent of element Means?enlist

Elements: the act (what is done) the means (how it is done) the purpose (why it is done)

Means: Threat, use of force, abduction, fraud, abuse of power, Payments/benefits

23) What is the **issue of human embryo stem cells and theoulptonic cloning. Explain? 5**

Human embryonic stem cells (hESCs): derived from embryo-5 to 7 days old-before implantation proliferate and differentiate hESCs generated organ face rejection by the immune system

Ethical issues: moral status of the embryo ---- destruction

patient has the right to live

morally right for in vitro fertilization but morally wrong to save a child's life

potential donor exploitation

slippery slope --------- reproductive cloning different from therapeutic cloning

no access and benefit to poor communities

raises issues of social justice and healthcare disparities

24)Write two types of Bio weapons? 2

chemical weapons, radiological weapons, nuclear weapons

25)What is the global warming effect. Explain any Five. 5

Earth climate zone is shifting, Polar ice start melting, Sea level increases, metabolic rate of methane producing bacteria increases, Species may extinct

Human activity: Transformation of forests in to lands

Pollution affect the ecosystem-loss of biodiversity

Tropical rain forest-climax ecosystem

Clearance of tropical forest: Use of wood

Need of a land

We are losing 7 million hectares per year Soil is degrading without trees

26)What is the biological reagents of bio weapons any six? 3

Anthrax, brucellosis, small pox, viral hemorrhagic fever, Staphylococaal enterotoxin B, botulinum toxins

27)Write three principle of bioethics? 3

Autonomy

nonmaleficence and beneficiance

Justice

28)Difference between sterilization and disinfection 5

Disinfection and sterilization are both decontamination processes. While disinfection is the process of eliminating or reducing harmful microorganisms from inanimate objects and surfaces, sterilization is the process of killing all microorganisms. That is the main difference between sterilizing and disinfecting.

29)What is Science in 21st century?

Science**:** Latin term "**Scio"** , observation and theoretical form, observation and experiment, investigation of the universe by a set of methodologies, progress made by scientific methods, Step-wise, not a single activity, not a value free.

Health care: technology/research/ education/administration/communication• couples can make their own decisions, organizational and global bioethics

30)What is Nanotechnology? Write its ethical issues?

Nanotechnology, also called *molecular manufacturing*, is "a branch of engineering that deals with the design and manufacture of extremely small electronic circuits and mechanical devices built at the molecular level of matter."The goal of nanotechnology is to be able to manipulate materials at the atomic level to build the smallest possible electromechanical devices, given the physical limitations of matter. Much of the mechanical systems we know how to build will be transferred to the molecular level as some atomic analogy. nanocomputers are no bigger than bacteria and nanomachines, also known as nanites, which could be used as a molecular assemblers and disassemblers to build, repair, or tear down any physical or biological objects.In essence, the purpose of developing nanotechnology is to have tools to work on the molecular level analogous to the tools we have at the macroworld level. Like the robots we use to build cars and the construction equipment we use to build skyscrapers, nanomachines will enable us to create a plethora of goods and increase our engineering abilities to the limits of the physical world.It would not take much of a leap, then, to imagine disassemblers dismantling garbage to be recycled at the molecular level, and then given to assemblers for them to build atomically perfect engines. Stretching this vision a bit, you can imagine a Star Trek type replicator which could reassemble matter in the form of a juicy steak, given the correct blueprints and organization of these nanomachines.Just given the basic premises of nanotechnology, you can imagine the vast potential of this technology. Some of its more prominent benefits would be:

Manufacturing

a. Precision Manufacturing
b. Material Reuse
c. Miniaturization

Medicine

d. Pharmaceutical Creation
e. Disease Treatment
f. Nanomachine-assisted Surgery

Environment

g. Toxin Cleanup
h. Recycling
i. Resource Consumption Reduction

Ethical issues: high reactivity and toxicity, distribution in the environment, ability to cross cell membranes and translocate in the body and economic effects/privacy issues

Or

Nanotechnology, also called *molecular manufacturing*, is "a branch of engineering that deals with the design and manufacture of extremely small electronic circuits and mechanical devices built at the molecular level of matter."The goal of nanotechnology is to be able to manipulate materials at the atomic level to build the smallest possible electromechanical devices, given the physical limitations of matter

Nanomedicine**:** nanorobots , injected in to the cancer patients-seek for cancerous cells, no side effects as of chemotherapy

Nanobiotechnology**:** cultured bladders, uterus grown outside the body, stem cell treatments, neurons can live together on a chip device

Bionanotechnology: DNA nanotechnology, chemical properties of lipids/proteins, build nanodevices with applications in engineering and medicine

Ethical issues: high reactivity and toxicity, distribution in the environment, ability to cross cell membranes and translocate in the body and economic effects/privacy issues

31)Write ethical issues cause by sybercrime?

Number of ethical issues has been raised like howmachines are in charge of key human functions. By this technique onlywealthy ones can communicate through cybernetics. Then whether implants are safe to use because senses and impulses can be transmitted in a harmful way? Can the senses be patented and who regulates?

Or

machines are in charge of key human functions
wealthy ones can communicate through cybernetics
implant are safe to use
senses and impulses –transmitted in a harmful way
can the senses be patented
who regulates?

32)Assessment of risks during Biotechnology practicals involved DNA and gene?

Rational and responsible assessment of risk requires that the following properties are all considered:

- source of the DNA of the target gene;
- source of the non target DNA segments of the construct used;
- site(s) of incorporation of the transgene within the recipient genome;
- product of the transgene;

- interaction of the transgenic product with other molecules in host and consumer;
- possible molecular changes in transgene product during processing;
- pleiotropic effects of transgene;
- tissue specificity of transgenic expression; and numbers of transgenic organisms capable of interacting with natural systems).

33)What is Exploitations responding to human trafficking victims?

Prostitution, sexual exploitation, forced labor, slavery, removal of organs

34)How biotechnology is misused

- reality of war
- inequitable distribution of resources
- frequent misuse of science act as constant reminder that our actions do not always live up to our aspirations

35)Enlist the types of ethical issues at the basis of scientific research

Meta ethics-----theoretical meaning

normative ethics-------practical meaning

applied ethics--domain of action

36)What is Aim of disaster management

- assistance to maintain life
- improve health
- support the morale- affected population
  - limited aid

37)Enlist the types of patient directives explain with examples

living will-applies to treatment such as dialysis –limited

- oral
- terminal illness— if patient die shortly
- health care power of attorney
- durable power of attorney
- agent/proxy
- agent make decisions

38)Human activity effect the biodiversity?

- Transformation of forests in to lands
- Pollution affect the ecosystem-loss of biodiversity
  - Tropical rain forest-climax ecosystem

39)What is Case study of spleen removal

Case study: A man presents with symptoms representing cancer and as part of his treatment spleen should be removed. The pathology department uses it to establish a cultured cell line in order to study the rare cancer. The cell line performs so well that the scientists collaborate with Biotechnology Company to patent it. They start to earn royalties from other laboratories and organizations that wish to use the cell line. When

patient find all this he was amazed, nobody has taken the consent from him nor he has been informed by anyone about these developments, analyze the ethical issues.
Reason: In terms of medical ethics, removal of the spleen was an act of doing good-beneficence. Patient's personal autonomy had been respected. It was for the sake of his health that spleen has been removed. In UK, there is great sensitivity concerning the fate of organ removed. Under new legislation, if there is no pre-death consent then kin permission must be taken to retain any organs from dead bodies. In USA, once the organ is removed during surgery, it is no longer belongs to the patient. **What is the purpose of keeping their appendix or diseased kidney in a jar in their office?** Donor cannot claim on the income gained as a result of research, who donates a kidney makes a gift not an investment in the recipient.

40)Difference between intrinsic and experimental values 5

Intrinsic value:

• value that environment and living forms have their own rights\

• intrinsic value of birds/green and pleasant places have their own values

• mainly involves religion

Instrumental value:

• Supply of human's material needs

• Actual and potential use in supplying resources for human living

• debate

41)What are Codes for nurse?

nurses and people

nurses and practice

nurses and profession

nurses and co-workers

42)why we use animals in bio technology?

to identify, isolate and characterize genes---- understand more about their function and

regulation, research models of human diseases, to provide organs and tissues , to produce milk

with therapeutic proteins or with improve nutritional values, to enhance livestock improvement programs

43)What is low risk in Tb labs?

The risk-assessment approach for a TB laboratory considers: the bacterial load of materials (such as sputum specimens and cultures), and the viability of TB bacilli; route of transmission of TB;

whether the material handled and the manipulations required for each procedure are likely to generate infectious aerosols;

the number of manoeuvres for each technique that may potentially generate aerosols;

the workload of the laboratory and individual staff members;
the location of the laboratory;
the epidemiology of the disease and the patient population served by the laboratory;
the level of experience and the competence of the laboratory's technicians;
the health of the laboratory's workers (especially HIV-positive technicians).
Or
Minimum requirements needed to limit or reduce risks of infection in laboratories by carrying out specific procedures that are considered to have a low risk of spreading TB. Additional measures may be deemed necessary following a site-specific risk assessment.Low-risk laboratories that follow the minimum biosafety requirements described in this chapter can safely perform certain procedures with sputum specimens, given that the viscous nature of sputum is not prone to generating aerosols when good microbiological techniques are followed. Low risk laboratories canmanipulate sputum specimens for direct sputum-smear microscopy; and manipulate sputum specimens for the Xpert MTB/RIF® assay.While opening sputum containers and making a direct sputum smear may produce aerosols, the risk of transmission from such procedures is negligible in comparison with aerosols produced by a single unprotected cough. There is little epidemiological evidence that preparing a direct smear is associated with a measurable excess risk of acquiring TB infection. The low-risk TB laboratory may also face the following challenges, all of which increase risks such as bench spaces may be used improperly;specimen containers may leak;specimens manipulated carelessly may lead to subsequent aerosolization;specimens may be shaken vigorously; and ventilation or illumination may be poor.
To address specific potential risks, the following biosafety requirements should be established in a low-risk TB laboratory.Use of bench spaces: The bench used to process specimens for direct sputum-smear microscopy or the Xpert MTB/RIF assay should be separate from areas used to receive specimens and from administrative areas used for paperwork and telephones.Ventilation: Smears performed directly on sputum samples, and processing specimens for the Xpert MTB/RIF assay, may both be carried out on an open bench in an adequately ventilated area when appropriate microbiological techniques are used

44)Define Trafficking

The movement of things illegally from one place to another there are many kinda of trafficking like drug trafficking, sex trafficking, goods trafficking andcommercial trafficking.

45)Define TO.R.C.H disease?

What is TORCH? TORCH is used for which diseases and why? Ans. Acronym TORCH is used for a blood test to screen out 4 infectious diseases

(Toxoplasmosis,Rubaila,cytomegaili,herpese).These diseases may transfer from mother to fetus and cause mental infections that lead to fetal mental retardation

46)Write about environment?
47)What is role of Biotechnology in health and medicine
48)Why personal health information should be kept confidential 5
49)What are fundamental moral principles of dealing with human embryos 5
50)How cybercrime is focused by information technology?
51)How human turns fresh water in swage?
52)What is the use uv-radiation in sterilizing?
53)Write exploitation of the victims of trafficking?
54) Why should patient cooperate with his\her caregiver? 2
55) What is the issue of human embryo stem cells and theoulptonic cloning. Explain ? 5
56)Difference between genetic screening and genetic testing?
57)Two manjor principal of how to treat embryo
58)What is the use of uv-radiation in sterifying
59)List the categories of hazards for a researcher while dealing with animals 3
60)What solution is injected when we inject a organ of hgm
61)How human turns fresh water in sewage
62)Write names of 10 biological spill kits
63)Write difference between objective bad news and subjective bad news?
64)Science is derived from a _____latin _____- word
65)Virtues is given by _____greek____ philosophers
66)The environment and living forms are involved in__anteinsio_-values
67)Which animal is now produced for fur? Mink
68)Aristotle idea rejected by_____ Thomas broke
69)Types of ethics_____ Meta ethics-----theoretical meaning, normative ethics-------practical meaning, applied ethics--domain of action
70)DNA consist of all except one_____
71)Patient has a right to_____ receive information, patient has a right to make decisions, patient has a right to confidentiality, Continuity--availability of health care
72)Informed consent_____ is legal and ethical right of the patient, permission before getting the healthca

Bt605 past papers for mid term